

Daúte Ricardo Novela

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PROFESSIONAL SUMMARY

Recently graduated Mechanical Engineer specialised in Machines and Technologies for High-Efficiency Material Processing, with strong applied knowledge in Materials Science, metallurgy, advanced surface engineering, and industrial quality assessment. Hands-on experience in evaluating material integrity and performance through laboratory and industrial methods, including laser-spark emission spectroscopy, metallography, ultrasonic testing, and Vickers hardness testing.

Experienced in translating engineering research into practical industrial solutions through process optimisation, technical analysis, and structured documentation, with results validated for industrial application. Strong understanding of quality assurance principles, materials evaluation, and technical problem-solving within demanding industrial environments.

Analytical, proactive, and adaptable engineer with proven ability to collaborate across multicultural teams and manage diverse technical challenges. Motivated to develop within the Testing, Inspection, and Certification industry, contributing to operational excellence while continuously expanding technical and leadership capabilities through a global graduate programme.

EDUCATION

National Research University "MPEI"

Sept 2021 – Jul 2025

Bachelor's Degree in Mechanical Engineering

Moscow, Russia

GPA: 4.2 / 5.0

Massinga Secondary School

Feb 2018 – Dec 2019

High School Diploma — Sciences

Inhambane, Mozambique

GPA: 18 / 20

INDUSTRIAL & RESEARCH EXPERIENCE

Engineering Industrial Practice | Plakart JSC

Jun 2024 – Jun 2025

Scherbinka, Moscow, Russia

- Applied laser cladding and surface hardening treatments to drilling column stabilisers, improving equipment durability and operational reliability under aggressive downhole conditions.
- Configured and operated laser cladding equipment, defining real-time process parameters for WC/nickel-based coatings on drilling column stabilisers; validated parameters were subsequently adopted for industrial application.
- Applied HS&E procedures during laser cladding complex operation, maintaining safe working conditions throughout industrial processing activities.

Materials Science Laboratory - Applied Research Practice | MPEI

Feb 2024 – May 2025

Krasnokazarmennaya 14, Moscow, Russia

- Conducted destructive and non-destructive evaluation of treated components to verify structural integrity, coating performance, and operational suitability.
- Investigated materials and laser cladding process parameters for drilling components, improving coating adhesion, wear resistance, durability, and operational performance under demanding downhole conditions.
- Authored a scientific article** on laser cladding of wear-resistant layers for drilling column stabilisers.

SKILLS & CORE COMPETENCIES

Technical Skills

- **Inspection, Analysis & NDT:** Laser-spark emission spectroscopy - LSES; Metallography; Vickers hardness testing; Ultrasonic testing; Coating performance assessment; Material integrity evaluation.
- **Manufacturing & Materials:** Laser Cladding; Additive Manufacturing; Surface Engineering; WC/Ni Superalloy Processing; Concentrated Energy Flow Technologies - laser, electron beam, plasma, induction; Welding Technologies - TIG, MIG, MMA.
- **Oilfield & Industrial Systems:** Equipment servicing; Preventive & corrective maintenance; Failure analysis; Drilling component integrity.
- **HS&E:** Safe equipment operation; Structural integrity assessment; Risk-critical component evaluation
- **Design & Simulation:** COMPASS-3D; Revit; Fusion 360; COMSOL Multiphysics; ELCUT.
- **Software & IT:** Microsoft Office; Industrial control software - laser systems.

Continuous Learning

- Python Programming — HarvardX CS50P, 2026 - Ongoing.
- PLC & Industrial Automation Fundamentals — Udemy, 2026 – Ongoing.

Professional Abilities

- Strong communication and teamwork in multicultural, international environments.
- Proactive, creative problem-solver — able to adapt rapidly to new systems and technologies.
- Adaptable to challenging field environments; comfortable working with industrial equipment and operational safety procedures.
- Experience collaborating with cross-functional teams including research, production, and quality assurance.
- Demonstrated leadership across academic and professional contexts.

Awards & Achievements

- International Medalist — CPLP Mathematics Olympiad, VIII Edition, São Tomé and Príncipe.
- National Medalist — Mozambique Mathematics Olympiad, Maputo, Mozambique.
- Honorary Diploma for Outstanding Academic Achievements - Governor of Inhambane Province

LANGUAGES

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| • English — Professional working proficiency | • Portuguese — Native or bilingual proficiency |
| • Russian — Full professional proficiency | • Tsonga — Native or bilingual proficiency |
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References and supporting documentation available upon request.

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